Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	Umai
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8: Decent Work and Economic Growth
Project / Solution Name	SurPrice
Name (Leader)	Ahmad Zhafir Bin Razali
Email & Phone (Leader)	zhafirrazali@gmail.com | +601112946874
University Affiliation(s)	Universiti Putra Malaysia
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small retail businesses, particularly mini marts and traditional grocery stores (kedai runcit), frequently face challenges in managing product prices when supplier costs change. Each time new stock arrives, retailers must manually review supplier invoices, identify affected products, calculate suitable selling prices, and update shelf price labels.

This repetitive process becomes increasingly difficult when retailers manage numerous products from different suppliers. Overlooked cost increases can reduce profit margins, while delayed shelf-label updates may cause inconsistencies between displayed and intended selling prices.

Additionally, certain essential goods in Malaysia are subject to government price controls. Retailers must consider applicable price ceilings when adjusting their selling prices, creating another layer of complexity in daily pricing decisions.

For micro and small retailers operating with limited manpower, these manual activities consume valuable time and increase the likelihood of pricing errors. Existing retail management solutions may also be too complex or costly for businesses that primarily need straightforward pricing assistance.

There is therefore a need for an accessible and affordable solution that simplifies supplier invoice processing, provides transparent pricing recommendations, supports compliance checks, and streamlines shelf-label updates.

Addressing this challenge can improve operational productivity, reduce repetitive administrative work, and support the digital transformation of Malaysian micro and small enterprises.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SurPrice is an AI-assisted retail pricing application designed to simplify the process of transforming supplier invoices into accurate, ready-to-print shelf price labels.

The solution follows a five-stage workflow: Capture, Understand, Propose, Approve, and Publish.

First, retailers upload or photograph a supplier invoice. SurPrice uses the Qwen2.5-VL-7B-Instruct vision-language model, deployed through a RunPod Serverless Endpoint, to extract supplier information, product names, quantities, units, and costs into structured JSON data.

Next, SurPrice validates the extracted information and matches invoice items against the retailer's existing product catalogue using product names, aliases, and similarity matching. It identifies supplier cost changes and generates recommended selling prices based on predefined profit-margin targets.

Unlike systems that rely entirely on AI-generated decisions, SurPrice uses deterministic pricing rules to calculate recommendations, flag significant cost increases, and check applicable price-control ceilings. This ensures pricing decisions remain explainable and verifiable.

Retailers review proposed prices before approval. Role-based authorisation ensures significant or flagged changes require the shop owner's approval. Once approved, product prices are updated and printable Bahasa Malaysia shelf labels are generated.

The prototype uses Python, FastAPI, SQLite, and a responsive web interface hosted on Replit, with RunPod providing GPU-powered AI inference. If AI reading fails, staff can enter the invoice lines manually.

SurPrice aims to improve pricing accuracy, reduce repetitive administrative work, and support the digital transformation of Malaysian micro and small retailers.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

SurPrice is developed as a web-based retail pricing application using a lightweight Python architecture. The deployed prototype integrates AI-powered invoice extraction, product matching, pricing recommendations, role-based approval management, and printable shelf-label generation.

The application is hosted on Replit, while AI inference is performed through a RunPod Serverless Endpoint running Qwen2.5-VL-7B-Instruct. Uploaded invoice images are sent securely to the endpoint, where the vision-language model extracts invoice information into structured JSON.

The extracted data is validated and matched against the retailer's product catalogue. Deterministic pricing rules calculate recommended prices, apply available price-control safeguards, and determine approval requirements.

Approved changes are recorded in SQLite, and printable Bahasa Malaysia shelf labels are generated. If live inference fails or times out, staff enter the invoice lines manually.

Proposed technologies and their functions

Component	Technology	Purpose
Frontend	HTML, CSS, Jinja2	Responsive user interface
Backend	Python, FastAPI	Application logic, routing, and business rules
Database	SQLite	Store product catalogues, invoices, pricing proposals, and audit records
AI model	Qwen2.5-VL-7B-Instruct	Extract structured information from invoice images
AI inference	RunPod Serverless Endpoint	GPU-powered model inference through REST API

Manual entry	Web form (manual line entry)	Handle failed or slow AI extraction
Product matching	Python, similarity matching	Match invoice items with existing catalogue products
Pricing engine	Python, deterministic rules	Calculate recommended prices and validate price ceilings
Authentication	Session-based authentication	Manage access by user role
Deployment	Replit	Host and demonstrate the web application
Output generation	HTML/CSS print layout	Generate printable Bahasa Malaysia shelf labels

Conceptual system block diagram



Official SDC sponsor-provided GenAI API or cloud platform:

SurPrice utilises Replit and RunPod as its cloud platforms for application deployment and AI inference.

Replit hosts the Python-based FastAPI application, providing browser-based access to invoice processing, pricing recommendations, approval workflows, and shelf-label generation.

RunPod Serverless Endpoints provide GPU-powered inference for the Qwen2.5-VL-7B-Instruct vision-language model. Supplier invoice images are transmitted from the FastAPI backend to the RunPod endpoint, which processes the images and returns structured invoice information in JSON format.

The integration has been implemented and evaluated using a synthetic invoice containing eight product lines. The team's GitHub README reports that all eight lines were extracted correctly in this controlled test.

This architecture separates AI-powered invoice interpretation from deterministic pricing decisions, ensuring that price calculations, compliance checks, and approvals remain controlled by the application.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

SurPrice adopts a design-thinking approach to address the daily pricing challenges faced by Malaysian micro-retailers.

During the Empathise and Define stages, the team identified repetitive invoice checking, manual price calculations, and inconsistent shelf-label updates as key operational challenges.

In the Ideate stage, the team developed a five-step workflow connecting invoice capture, AI-powered extraction, pricing recommendations, human approval, and shelf-label generation.

During Prototyping, the team implemented a functional web application using FastAPI and SQLite, deployed on Replit. AI invoice extraction was integrated through a RunPod Serverless Endpoint running Qwen2.5-VL-7B-Instruct.

Preliminary technical testing includes 13 synthetic rule-breach scenarios, all successfully intercepted by the pricing engine. An initial RunPod inference test also extracted all eight invoice lines correctly from one synthetic sample. These results demonstrate the functionality of the prototype but do not establish accuracy across real-world invoice formats.

For Phase 2, the team will focus on testing diverse real supplier invoices, evaluating AI extraction accuracy and processing time, conducting user task-completion tests, and improving deployment reliability.

The deployment strategy prioritises a lightweight, browser-accessible solution without requiring specialised hardware from retailers.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG Target 8.2: Improve economic productivity through technological upgrading and innovation. SDG Target 8.3: Support productive activities, entrepreneurship, innovation, and the growth of micro, small and medium enterprises.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	No measured manual-processing baseline is available yet. For a preliminary controlled test, assume 10 minutes to manually review a 15-item invoice, calculate price changes, and prepare updated shelf labels. This figure is a testing assumption, not an observed result.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Average invoice-to-label processing time (minutes per 15-item invoice), measured from invoice review to approved, printable shelf labels.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Target at least a 40% reduction in processing time compared with the preliminary 10-minute baseline, aiming to complete the same workflow in 6 minutes or less.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
FastAPI	Open-source code framework	MIT
Starlette	Open-source code framework	BSD-3-Clause
Uvicorn	Open-source code framework	BSD-3-Clause
Jinja2	Open-source code framework	BSD-3-Clause
SQLite	Database engine	Public domain
Pillow	Open-source code library	HPND
httpx	Open-source code library	BSD-3-Clause
Python	Programming language/runtime	PSF License
Qwen2.5-VL-7B-Instruct	Pretrained vision-language model	Apache-2.0
RunPod Serverless	Cloud inference platform/API	RunPod service terms
RunPod Python SDK	Open-source library	Verify installed version's license
PyTorch	Machine-learning framework	BSD-3-Clause
Transformers	Model inference library	Apache-2.0
Accelerate	Model execution library	Apache-2.0

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



Ahmad Zhafir
24/10/26